# Lab 1

NO late submission. Grade = 0 after the deadline.

**Due**: 13:00h 03/04/2026

- Submission:
    - **Lab#_YOURNAME.ipynb**
    - Must include all data file (**.csv,…)** and **.ipynb** in your submission
    - **.csv** and **.ipynb** must be in the **same** folder (pd.read_csv('data.csv'))
    - No use of LOCAL libraries, NO local directory for libraries

Use '**Lab1_dataset.csv**'.

a) Missing data could be anything in { NA, n/a, " ", -1, …}. Read the data and assign the missing data in the dataframe as **NaN**. Hint: df.***replace***( …, numpy.nan, inplace=True)

b) Show the summary of the dataframe

c) Count the ***number*** and ***percentage*** of rows with missing in the entire dataset

d) Count the ***number*** and ***percentage*** of missing data in each

e) ***Randomly*** drop 10% of the rows that have missing data. Hint: use **any()** and **numpy.random.choice()**

f) Fill in the missing data with ***median*** (if the column data type is numeric) or ***mode*** (if the column data type is not numeric)

g) Show the summary of the ***updated*** dataframe

h) Extract the row with ***max*** SalePrice and the row with ***min*** SalePrice

i) Insert a new column called ***Overall***, which is the average of OverallQual and OverallCond.

j) Update the dataframe with SalePrice used as ***index*** and ***ID*** becomes a new column